

SSH 目录不可创建文件问题修复

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1 问题发现

最开始发现问题是使用 apt 更新软件

```
wpc@dfs-server-02:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
You might want to run 'apt --fix-broken install' to correct these.
The following packages have unmet dependencies:
 openssh-sftp-server : Depends: openssh-client (= 1:8.9p1-3ubuntu0.11)
E: Unmet dependencies. Try 'apt --fix-broken install' with no packages (or specify a solution).
```

尝试运行 apt --fix-broken install 修复问题

```
wpc@dfs-server-02:~$ sudo apt --fix-broken install
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Correcting dependencies... Done
The following additional packages will be installed:
 openssh-client openssh-server
Suggested packages:
 keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following packages will be upgraded:
 openssh-client openssh-server
2 upgraded, 0 newly installed, 0 to remove and 134 not upgraded.
1 not fully installed or removed.
Need to get 0 B/1,337 kB of archives.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] y
Preconfiguring packages ...
(Reading database ... 242404 files and directories currently installed.)
Preparing to unpack .../openssh-server_1%3a8.9p1-3ubuntu0.11_amd64.deb ...
Unpacking openssh-server (1:8.9p1-3ubuntu0.11) over (1:8.9p1-3ubuntu0.10) ...
dpkg: error processing archive /var/cache/apt/archives/openssh-server_1%3a8.9p1-3ubuntu0.11_amd64.deb (--unpack):
 unable to create '/etc/ssh/moduli.dpkg-new' (while processing './etc/ssh/moduli'): Operation not permitted
dpkg-deb: error: paste subprocess was killed by signal (Broken pipe)
rescue-ssh.target is a disabled or a static unit not running, not starting it.
ssh.socket is a disabled or a static unit not running, not starting it.
Preparing to unpack .../openssh-client_1%3a8.9p1-3ubuntu0.11_amd64.deb ...
Unpacking openssh-client (1:8.9p1-3ubuntu0.11) over (1:8.9p1-3ubuntu0.10) ...
dpkg: error processing archive /var/cache/apt/archives/openssh-client_1%3a8.9p1-3ubuntu0.11_amd64.deb (--unpack):
 unable to create '/etc/ssh/ssh_config.dpkg-new' (while processing './etc/ssh/ssh_config'): Operation not permitted
dpkg-deb: error: paste subprocess was killed by signal (Broken pipe)
Errors were encountered while processing:
 /var/cache/apt/archives/openssh-server_1%3a8.9p1-3ubuntu0.11_amd64.deb
```

2 着手修复

观察控制台输出，可知是 openssh-server 和 openssh-client 两个软件包的问题，尝试重装这两个软件包

1. 卸载包:

```
sudo apt remove openssh-server openssh-client --purge
sudo apt autoremove
sudo apt autoclean
```

2. 仍有 error, 强制删除:

```
sudo dpkg -P --force-all openssh-server openssh-client
```

3. 尝试修复:

```
sudo apt-get install -f
```

经过上述步骤, 问题仍没解决, 不过输出的 error 少了许多, openssh-server 的问题已经解决, 但 openssh-client 仍有问题

```
dpkg: error processing archive /var/cache/apt/archives/openssh-client_1%3a8.9p1-3ubuntu0.11_amd64.deb (--unpack):
unable to create '/etc/ssh/ssh_config.dpkg-new' (while processing './etc/ssh/ss
h_config'): Operation not permitted
```

3 问题分析与修复

从控制台输出看出问题出在无法创建文件上, 尝试手动创建文件

```
cd /etc/ssh
touch ssh_config.dpkg-new
```

出现 error, touch: setting times of 'xxx': No such file or directory

使用 ls 查看 /etc/ssh 目录权限, 并未发现问题,

```
wpc@dfs-server-02:/etc$ ls -al | grep ssh
drwxr-xr-x  4 root  root    4096  3月 25 11:30 ssh
```

估计是设置了隐藏属性, 使用 lsattr 查看目录属性, 发现目录被锁定 使用 chattr -ia ./ssh 修复问题, 到此问题解决

```
cedfs-server-02:/etc$ lsattr | grep ssh
attr: Permission denied While reading flags on ./cni
attr: Permission denied While reading flags on ./shadow-
attr: Permission denied While reading flags on ./sudoers
attr: Permission denied While reading flags on ./chatscripts
attr: Operation not supported While reading flags on ./resolv.conf
attr: Operation not supported While reading flags on ./printcap
attr: Operation not supported While reading flags on ./localtime
attr: Operation not supported While reading flags on ./rmt
attr: Operation not supported While reading flags on ./os-release
attr: Permission denied While reading flags on ./gshadow
attr: Permission denied While reading flags on ./shadow
attr: Operation not supported While reading flags on ./vtrgb
attr: Operation not supported While reading flags on ./mtab
attr: Permission denied While reading flags on ./gshadow-
--ls-----e----- ./ssh
attr: Permission denied While reading flags on ./brapi.key
cedfs-server-02:/etc$
```